TEACHER GUIDE

ANSWERS AND MARKS

GRADE 9 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 3
Programming



(!) FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 3 Student Workbook, in the same order. The exercise letters match the workbook.

(i) WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - what to check in Practicals 1 - 4, an oral quiz, a practical test
- **Answers** - chapters 1 to 4, in the same order as the workbook
- **Unit 3 Test** - the marking scheme and the common mistakes
- **Extra** - textbook links, homework ideas and teaching tips

(★) BEFORE THE FIRST LESSON

- ☐ Scratch opens on every computer (the offline editor or scratch.mit.edu).
- ☐ You have a coin, a dice and a Snakes and Ladders board for the class.
- ☐ The projector works, so you can build each program with the class first.

"Let them make the computer do something. Children learn programming by building it."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A Complete the flowchart	6
	B Fill in the blanks	6
	C What will the cat say?	6
	D Answer these	4
	E Complete the script	5
	F Simple or many options?	6
	G True or false	6
Chapter 2	A Follow the flowchart	8
	B Complete the flowchart	6
	C What will the cat say?	6
	D Mark the register	6
	E Which repeat block?	6
	F True or false	6
Chapter 3	A Follow the program	8
	B Count the repetitions	6
	C Play the game	8
	D True or false	6
	E Answer these	4
Chapter 4	A Read the lists	6
	B What does each block do?	6
	C Who is selected?	6
	D Follow the average program	6
	E Sections of a problem	8
	F Crossword	10
End	Unit 3 Test	40
TOTAL		**191**

(!) SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

(★) TEACHING TIP

- Ask: why is there no "Is value 6?" in Flowchart 2? It is the best question of 3.2.

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (39)	Ch 2 (38)	Ch 3 (32)	Ch 4 (42)	Test (40)	Total (191)
1							
2							
3							
4							
5							
6							
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28							
29							

TEACHER GUIDE

Practicals at a Glance

Workbook activities 3.1 to 3.4 (textbook pages 31 - 46)

Practical	Students make	Check that ...
1 (Activity 3.1)	the head or tail program (Scratch program 1)	Side landed is set to answer; Head gives "Head" and Tail (or anything else) gives "Tails".
2 (Activity 3.2)	Scratch programs 3, 4 and 5 with repeat, repeat until and forever	program 3 says 1 - 10 in 10 seconds; program 4 stops after 5; program 5 is stopped with the red stop button.
3 (Activity 3.3)	Scratch program 10 (the squares) and the Snakes and Ladders game (Scratch program 11)	the repeat 4 block is inside repeat 100; the game asks the three questions in every turn and says "Game is over" at square 36 or more.
4 (Activity 3.4)	the Country list (Scratch programs 13 and 14) and the average program (Scratch program 22)	the list shows 5 countries; v_con starts at 1 so all five are said; the average agrees with a calculator.

! MATERIALS

- Scratch on every computer (the offline editor, or scratch.mit.edu with the Internet). The textbook shows Scratch 2: lists and variables are in Data (in Scratch 3: Variables).
- A coin, a dice and a Snakes and Ladders board for each group (Practicals 1 and 3).
- A projector, so that you can build each program with the class first.
- The textbook points to Activities 3.2 and 3.3 at the end of 3.6 (page 39). In this workbook Practical 2 (Activity 3.2) comes after 3.4, where its three blocks are taught.

Textbook links

Workbook	Textbook pages	Activities
Chapter 1 - Selection (3.1, 3.2)	29 - 32	3.1
Chapter 2 - Repetition (3.3, 3.4)	32 - 34	3.2
Chapter 3 - Programs with repetition (3.5, 3.6)	35 - 39	3.3
Chapter 4 - Arrays and problem analysis (3.7, 3.8)	40 - 46	3.4
Unit 3 in One Page	46 (summary)	-

★ TEACHING TIPS

- Toss a real coin and a real dice in class before you open Scratch. Let the class say what the program must do.
- Act out Flowchart 2: one student holds a dice, the class asks "Is the value 1?", "Is the value 2?" ...
- Draw the loop-back arrow of a flowchart in a different colour on the board. Students see the repetition at once.
- Trace Flowchart 3 on the board with a table: Number, condition, shown. Use the same table for every loop.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. One condition, two options: what kind of selection?	**Simple selection**
2. How many decision symbols does the dice flowchart have?	**Five**
3. An action repeated again and again is called ...	**Repetition**
4. Which block repeats a definite number of times?	**repeat**
5. Which block repeats till a condition is satisfied?	**repeat until**
6. Which block never stops?	**forever**
7. A repetition within a repetition is called ...	**Nested repetition**
8. repeat 4 inside repeat 100: how many moves?	**400**
9. What is an array called in Scratch?	**A list**
10. Which list block gives the number of items?	**length of**

> PRACTICAL TEST - 20 MINUTES, 20 MARKS (OPTIONAL)

In the computer room after the unit. The student builds the first 10 multiples of a number (like Scratch program 8). Read the steps aloud; mark on the screen.

Step	The student must ...	Marks
1	Make a variable **Count** and start with **when green flag clicked**.	2
2	**ask** Enter a number and wait; **set Number to answer**.	3
3	**set Count to 1** and add **repeat until Count > 10**.	4
4	Inside: **say Number * Count for 1 secs** and **change Count by 1**.	5
5	Test it with 4: the cat says 4, 8, 12 ... 40.	3
6	Save the project with your name.	3
	Total	**20**

ANSWERS

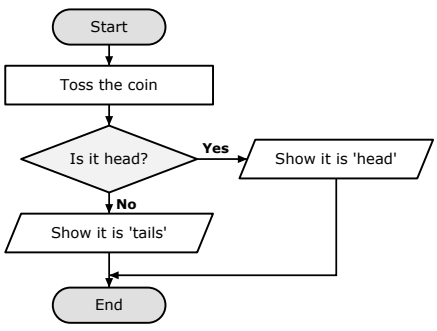
Chapter 1

Selection

A Complete the flowchart

6 marks

1. **Toss the coin**
2. **Is it head?**
3. **Show it is 'head'**
4. **Show it is 'tails'**
5. **Yes**
6. **No**



1 mark each. 5 is the Yes arrow, 6 the No arrow.

B Fill in the blanks

6 marks

1. two	**2.** one	**3.** more than two
4. true	**5.** else	**6.** decision

1 mark each. The extra words are **repeat** and **six**.

C What will the cat say?

6 marks

1. Program 1, Head: **Head**	**2.** Program 1, Tail: **Tails**
3. Program 2, 2: **2**	**4.** Program 2, 5: **5**
5. Program 2, 6: **6**	**6.** Program 2, 9: **6**

Tail is not equal to Head, so the else part runs: **Tails**. 9 is not 1, 2, 3, 4 or 5, so none of the five conditions is satisfied and the last else part says **6**. 1 mark each.

D Answer these

4 marks

1. What is the difference between simple selection and selection out of many options? (2 marks)

Simple selection selects one option out of two (one condition, two options). Selection out of many options selects one option out of more than two options, so several conditions are checked. 2 marks.

2. Scratch program 2 does not check "Value = 6". Why is it not needed? (2 marks)

When the value is not 1, 2, 3, 4 or 5, none of the five conditions is satisfied, so the value must be 6: the last else part shows 6. 2 marks.

E Complete the script

5 marks

1. Toss the coin	**2.** answer	**3.** =	**4.** Head	**5.** Tails

1 mark each. The extra words are **Value** and **>**.

TEACHING TIPS

- Count with the class while Scratch program 3 runs: 1, 2, 3 ... 10 in 10 seconds.
- Show that repeat until checks the condition first: set A to 10 and repeat until A > 5 says nothing.
- Let students stop the forever program with the red stop button. They must know how to stop a program.
- For multiples, let students say the 7 times table aloud while Scratch program 8 runs.
- Scratch program 10 draws for some time. Let students guess the picture before it finishes.

F Simple or many options?

6 marks

Selection	Simple selection	Many options
Head or tail when you toss a coin	☑	☐
The value of a tossed dice: 1, 2, 3, 4, 5 or 6	☐	☑
One condition with two options	☑	☐
Selecting one option out of more than two options	☐	☑
Is the student present? Mark 1 or mark 0.	☑	☐
Five IF THEN ELSE blocks, one inside another	☐	☑

1 mark each.

G True or false

6 marks

1. **T - Simple selection has one condition with two options.**
2. **F - In Flowchart 1, when the coin shows head, the computer shows "tails".**
3. **T - Selection out of many options selects one option out of more than two.**
4. **F - Scratch program 2 has six IF THEN ELSE blocks.**
5. **T - In Scratch program 2, the answer is kept in the variable Value.**
6. **T - When none of the five conditions is satisfied, the dice program shows 6.**

Number 4: there are **five** IF THEN ELSE blocks (Value = 1 ... Value = 5).

TEACHING TIP

- Revise the word list with a quick game: you read a meaning, the class says the word.

ANSWERS

Chapter 2

Repetition

A Follow the flowchart

8 marks

Number	5 or less than 5?	What is shown?
1	Yes	1
2	Yes	2
3	Yes	3
4	Yes	4
5	Yes	5
6	No	nothing (End)

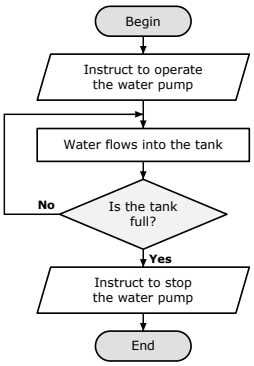
1. How many numbers are shown? **5**
2. How many times is the condition checked? **6**
3. What is the value of Number when the repetition ends? **6**

1 mark for each of the rows 2 - 6 and for each question. Number 6: the condition is false, so nothing is shown and the repetition ends.

B Complete the flowchart

6 marks

1. **Instruct to operate the water pump**
2. **Water flows into the tank**
3. **Is the tank full?**
4. **Instruct to stop the water pump**
5. **No**
6. **Yes**



1 mark each. 5 (No) loops back up; 6 (Yes) goes down to the stop.

C What will the cat say?

6 marks

1. Script 1: **1, 2, 3**
2. Script 2: **2, 4, 6, 8**
3. Script 3: **Nothing. 10 > 5 is already true, so the blocks inside never run.**

2 marks each. Script 3: repeat until checks the condition first.

D Mark the register

6 marks

1. Kasun: **1**	**2.** Dilini: **0**	**3.** Amal: **1**	**4.** Nimal: **0**	**5.** Sachini: **1**

Present: 1, absent: 0. Question 1: **4** times. After the fifth name there are no more names, so the No arrow is followed once and the Yes arrow 4 times. 1 mark each.

E Which repeat block?

6 marks

1. Say the numbers 1 to 10, one number each second. → **repeat**
2. Say the numbers from 1 until A is greater than 5. → **repeat until**
3. Say all the positive integers from 1, without stopping. → **forever**
4. Ask for the names of five countries. → **repeat**
5. Show a random number from 1 to 100 again and again, without end. → **forever**
6. Show the multiples of a number until Count is greater than 12. → **repeat until**

F True or false

6 marks

1. **T - Repetition is about an action getting repeated again and again.**
2. **F - A repetition does not need a condition to begin or to end.**
3. **T - The repeat block repeats its instructions a definite number of times.**
4. **F - The forever block stops when A is greater than 5.**
5. **T - In Flowchart 4, the water pump stops when the tank is full.**
6. **F - In Flowchart 3, 1 is added to the number before the number is shown.**

ANSWERS

Chapter 3

Programs with repetition

A Follow the program

8 marks

Count	Count > 12?	The cat says
1	No	7
2	No	14
3	No	21
4	No	28
...	...	...
12	No	84
13	Yes	nothing (it ends)

1. How many numbers does the cat say? **12**
2. What is the value of Count when the repetition ends? **13**
3. Scratch program 6 gives the same output. Which repetition block does it use? **repeat (12)**

The cat says the 12 multiples of 7: 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84. 1 mark for each of the rows 2, 3, 4, 12 and 13 and for each question.

B Count the repetitions

6 marks

1. How many times does the outer repeat block run? **100**
2. How many times does the inner repeat block run, each time? **4**
3. How many times does the **move 100 steps** block run altogether? **400**
4. How many degrees does the sprite turn to draw one square? **360 (4 × 90)**
5. How many squares does the program draw? **100**
6. Which block gives each square a new colour? **change pen color by 10**

C Play the game

8 marks

Turn	Dice	Working	Square
1	3	0 + 3 = 3	**3**
2	2	3 + 2 = 5, ladder up to 14	**14**
3	4	14 + 4 = 18, ladder up to 29	**29**
4	4	29 + 4 = 33, snake down to 20	**20**
5	6	20 + 6 = 26	**26**
6	5	26 + 5 = 31	**31**
7	6	31 + 6 = 37	**37**

Question 1: **Game is over**. After turn 7 the square is 37: 37 > 36 is true, so the repetition ends. 1 mark for each square and 1 for the question.

TEACHING TIP

- Count nested repetition with your hands: 4 sides in one square, 100 squares, so 400 moves.

D True or false

6 marks

1. **T - A repetition within a repetition is called a nested repetition.**
2. **F - In Flowchart 7, the condition "Is number > 0?" changes to false after some time.**
3. **T - Scratch program 6 and Scratch program 8 give the same output.**
4. **T - In Scratch program 10, the repeat 4 block is inside the repeat 100 block.**
5. **F - Scratch program 9 stops after it draws 360 lines.**
6. **T - The Snakes and Ladders program repeats until the number of the square is 36 or more.**

E Answer these

4 marks

1. Why does the repetition in Flowchart 7 never end? (2 marks)

The random number is always from 1 to 100, so "Is number > 0?" is always true. It never changes to false, so the repetition goes on and on (continuous repetition). 2 marks.

2. Scratch program 11 has repetition and selection. Write its repetition block and one selection block. (2 marks)

Repetition: repeat until (number of the square > 36 or = 36). Selection: if ... then ... else (or if ... then). 1 mark each.

ANSWERS

Chapter 4

Arrays and problem analysis

A Read the lists

6 marks

1. item 3 of Name gives **Radha**
2. item 5 of Marks gives **60**
3. length of Marks gives **5**
4. Name contains Saman? gives **true**
5. Name contains Kamal? gives **false**
6. The mark of Pawani is **65**

B What does each block do?

6 marks

1. add thing to Animal → **C**	**2.** delete 1 of Animal → **E**
3. insert thing at 1 of Animal → **B**	**4.** item 1 of Animal → **F**
5. length of Animal → **A**	**6.** Animal contains thing ? → **D**

TIP: Ask students to explain a program in their own words before they build it.

C Who is selected?

6 marks

1. Nayana (74): **Nayana Not selected**	**2.** Pawani (65): **Pawani Not selected**
3. Radha (87): **Radha Selected**	**4.** Paushi (42): **Paushi Not selected**
5. Saman (60): **Saman Not selected**	

Only Radha has more than 75. The join block puts the name and the word together (RadhaSelected); accept it with or without a space. Question 1: **No. 75 > 75 is false: only marks greater than 75 are selected.** 1 mark each.

D Follow the average program

6 marks

1. The value of total after the first repetition **4**	**2.** The value of total after the third repetition **16**
3. The value of total at the end **60**	**4.** The value of count at the end **11**
5. The value of average **6**	**6.** The number of times the ask block runs **10**

The totals after each repetition: 4, 11, 16, 24, 30, 39, 42, 49, 54, 60. Average = 60 / 10 = 6. 1 mark each.

E Sections of a problem

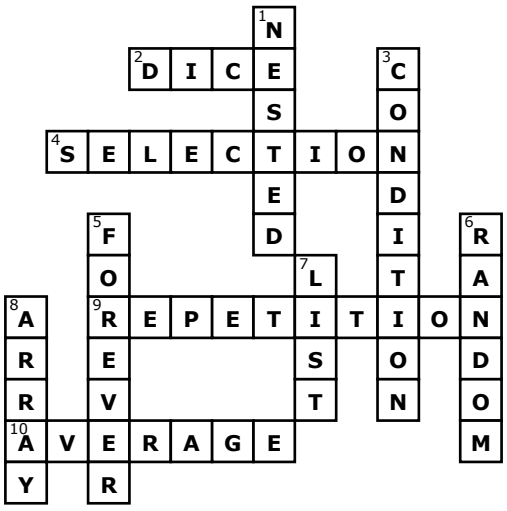
8 marks

Section	Order	Scratch program
Divide the total by ten to find the average.	**3**	**20**
Output the average.	**4**	**21**
Input ten numbers.	**1**	**18**
Find the total of the ten numbers.	**2**	**19**

1 mark for each order number and 1 for each program number.

F Crossword

10 marks



Across: 2 DICE, 4 SELECTION, 9 REPETITION, 10 AVERAGE

Down: 1 NESTED, 3 CONDITION, 5 FOREVER, 6 RANDOM, 7 LIST, 8 ARRAY

MARKING SCHEME

Unit 3 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (b) selection	**2.** (c) 5
3. (b) repetition	**4.** (b) full
5. (b) 1	**6.** (d) 36
7. (b) circle	**8.** (b) easier

2 Fill in the blanks

8 marks (1 each)

1. two	**2.** decision	**3.** condition	**4.** 10
5. true	**6.** array	**7.** list	**8.** average

3 What does each block do?

8 marks (1 each)

1. Repeats the blocks inside it 10 times (a definite number of times).
2. Repeats the blocks inside it till the condition is satisfied (becomes true).
3. Repeats the blocks inside it an infinite number of times (it never stops).
4. Runs the blocks inside it only when the condition is true (selection).
5. Adds the item (thing) at the end of the list Name.
6. Deletes item 1 of the list Name.
7. Gives item 1 of the list Name.
8. Gives the number of items in the list Name.

Accept any correct short sentence.

4 Answer these

16 marks

1. How is simple selection different from selection out of many options? (2 marks)

Simple selection: one condition, one option out of two (head or tail). Selection out of many options: one option out of more than two, several conditions (the dice). 1 mark each.

2. Look at the script. Write what the cat says. How many times does it repeat? (3 marks)

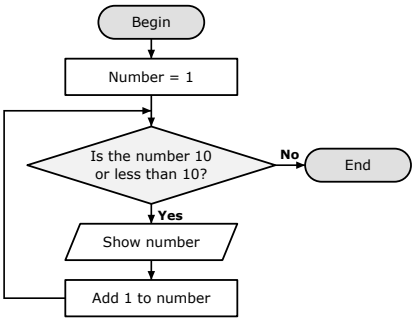
The cat says 3, 4, 5, 6 (2 marks). It repeats 4 times (1 mark). When A becomes 7, 7 > 6 is true and the repetition ends.

3. What is an array? Write two advantages of using arrays. (3 marks)

A data structure that can store any number of items using a single variable name (1 mark). Any two (1 mark each): no need to name many variables; the program becomes less complex; fewer instructions; a repetition can be used to handle all the items.

4. Draw a flowchart to display the numbers 1 to 10. (5 marks)

1 mark each: Begin and End; Number = 1; the decision (Is the number 10 or less than 10?) with Yes and No; Show number; Add 1 to number with the arrow back to the decision. Accept other correct flowcharts.



5. A repeat 4 block with a move block in it is inside a repeat 100 block. (a) What is this called? (b) How many times does the move block run? (3 marks)

(a) A nested repetition (a repetition within a repetition) - 1 mark. (b) 100 × 4 = 400 times - 2 marks (1 for the working).

WHAT STUDENTS USUALLY GET WRONG

- They think Scratch program 2 needs a sixth condition for 6. The last else part shows 6.
- They show the number in Flowchart 3 before checking the condition. The condition comes first.
- They think repeat until stops when the condition is false. It stops when the condition becomes **true**.
- They add the counts of a nested repetition (100 + 4) instead of multiplying them (100 × 4 = 400).
- They forget that the forever block never stops by itself: click the red stop button.
- They start a list at item 0. In Scratch the first item is item 1.
- They think 75 > 75 is true. Equal is not greater, so a mark of 75 is not selected.

HOMEWORK AND EXTRA PRACTICE

- Draw a flowchart that displays the even numbers 2, 4, 6 ... 20.
- Write a trace table for Scratch program 8 with your own number.
- Make a list of five friends and their birthdays. Which list blocks would you use?
- Fast learners: change Scratch program 10 to draw triangles (repeat 3, turn 120 degrees).

TEACHING TIP

- Play a real game of Snakes and Ladders in pairs while Scratch program 11 asks the questions.

WHERE ARE THE BLOCKS? - SCRATCH 2 AND SCRATCH 3

The textbook pictures show Scratch 2. The blocks are the same in Scratch 3, but some groups have new names:

Blocks of this unit	Scratch 2 (textbook)	Scratch 3
repeat, repeat until, forever, if then else, wait	Control	Control
set, change, Make a Variable	Data	Variables
Make a List, add, delete, insert, item, length of	Data	Variables
ask and wait, answer	Sensing	Sensing
say, say for 2 secs	Looks	Looks
join, pick random, +, *, /, =, >, or	Operators	Operators
go to x: y:, move, turn	Motion	Motion
clear, pen up, pen down, change pen color	Pen	Pen (an extension: add it first)

EXPECTED RESULTS OF THE PRACTICALS

Practical	On the screen the student should see ...
1	Type Head: the cat says "Head". Type Tail (or any other word): the cat says "Tails".
2	Program 3: 1, 2, 3 ... 10, one number each second (10 seconds). Program 4: 1 - 5, then it stops (A > 5 is true). Program 5: 1, 2, 3 ... it never stops - click the red stop button.
3	Program 10: 100 coloured squares round the centre (Figure 3.3). The game asks three questions in every turn and says "Game is over" when the square is 36 or more.
4	The Country list on the stage has 5 items and the cat says all five. Program 22 says the average; it agrees with a calculator (the total divided by 10).

TEACHING TIPS

- Scratch 2 calls the variables and lists group "Data". In Scratch 3 it is "Variables". Show both if you can.
- Show the list on the stage while Scratch program 13 runs. Students see the array grow item by item.
- Set v_con to 1 in Scratch program 14. A Scratch list starts at item 1; with 0 the first say is empty.
- Keep a row of five boxes (marks_1 ... marks_5) on the board next to one long box called Marks.
- Ask the class to check the average of Scratch program 22 with a calculator, then change one number.
- Problem analysis: write the four sections on four cards and let students put them in order.
- Pairs: one student reads a program aloud block by block, the other writes what the cat says.